EcoSort: Smart Waste Management System

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***Abstract*—Waste management has become a major environmental challenge due to increasing urbanization and population growth. Improper segregation of waste at the source leads to inefficient recycling and increased landfill usage. Traditional waste segregation methods rely heavily on manual sorting, which is time-consuming and exposes workers to hazardous materials.**

This paper presents EcoSort, a smart waste management system that automatically classifies waste into metallic, wet, and dry categories using computer vision and machine learning techniques. The system uses a webcam paired with the YOLOv11 object detection model to identify waste objects placed into the bin. Based on the detected waste category, a microcontroller-controlled sorting mechanism directs the waste into the appropriate compartment.

The proposed system aims to improve waste segregation efficiency while reducing human intervention. By combining real-time object detection with automated mechanical sorting, EcoSort demonstrates a practical approach toward smart waste management solutions.

I. Introduction

Waste generation has increased significantly due to rapid urbanization, population growth, and rising consumption patterns. Improper waste management practices have led to severe environmental issues such as pollution, landfill overflow, and inefficient recycling processes.

One of the key challenges in modern waste management systems is the lack of segregation at the source. In many cases, different types of waste such as organic waste, recyclable materials, and metallic objects are disposed of together in a single container. This mixed waste makes recycling difficult and reduces the efficiency of waste processing systems.

Recent advancements in **computer vision and machine learning** have enabled automated systems capable of identifying objects in real time. Object detection models such as **YOLO (You Only Look Once)** can accurately classify objects from images captured by cameras.

The **EcoSort system** utilizes a webcam combined with a YOLO-based detection model to identify different types of waste and automatically segregate them into appropriate compartments. This approach enables intelligent waste classification and contributes to more efficient waste management practices.

II. Problem Definition

Improper waste segregation remains a major problem in waste management systems. When different types of waste are mixed together, recyclable materials become contaminated and cannot be processed effectively. This leads to increased landfill waste and environmental pollution.

Manual segregation of waste is labor-intensive and inefficient, particularly in large urban areas where waste generation is high. Additionally, sanitation workers are exposed to potentially hazardous materials while manually sorting waste.

Existing automated waste segregation systems often rely on simple sensors to detect waste characteristics such as moisture or metal presence. However, these systems have limitations in accurately identifying different types of waste materials.

To address this problem, the **EcoSort system proposes a computer vision based waste classification approach**. The system uses a **webcam to capture images of the waste item** placed in the bin. These images are processed using a **YOLOv11 object detection model**, which identifies the type of waste.

Once the waste category is detected, the system activates a **motor-driven sorting mechanism** that directs the waste into the appropriate compartment for **metallic, wet, or dry waste**.

This approach enables intelligent and automated waste segregation using real-time image processing and machine learning techniques.

III. Methodology

The EcoSort system combines **computer vision, embedded systems, and mechanical automation** to perform intelligent waste segregation. The methodology involves capturing images of waste items, classifying them using a trained object detection model, and directing them into designated compartments using a motorized sorting mechanism.

A. System Overview

The EcoSort system is designed as an automated waste segregation unit that integrates **computer vision, embedded control, and mechanical sorting mechanisms**. The system is composed of three main subsystems: the image acquisition unit, the waste classification unit, and the mechanical sorting unit.

A **webcam** is positioned near the waste input slot to capture images of the waste object when it is inserted. The captured image is processed using the **YOLOv11 object detection model**, which identifies the type of waste placed in the bin. The model classifies the waste into three categories: **metallic waste, wet waste, and dry waste**.

Once the waste category is identified, the control system activates the sorting mechanism to direct the waste into the appropriate compartment. The hardware components used in the EcoSort system include:

- Arduino Uno microcontroller
- Rain drop sensor (wet waste detection)
- Proximity sensor
- IR sensor
- Servo motor
- Stepper motor
- Segmented waste bin

The **Arduino Uno microcontroller** acts as the central control unit that coordinates sensor inputs and motor outputs. The rain drop sensor assists in detecting moisture content in waste, while the proximity and IR sensors help detect the presence of waste objects.

The mechanical sorting mechanism consists of a **stepper motor-driven rotating platform** and a **servo motor-controlled flap gate**. Based on the detected waste category, the platform rotates to align the appropriate compartment, and the flap gate releases the waste into the designated segment within the bin.

This integration of computer vision, sensors, and motor-driven mechanisms enables the EcoSort system to perform automated and efficient waste segregation.

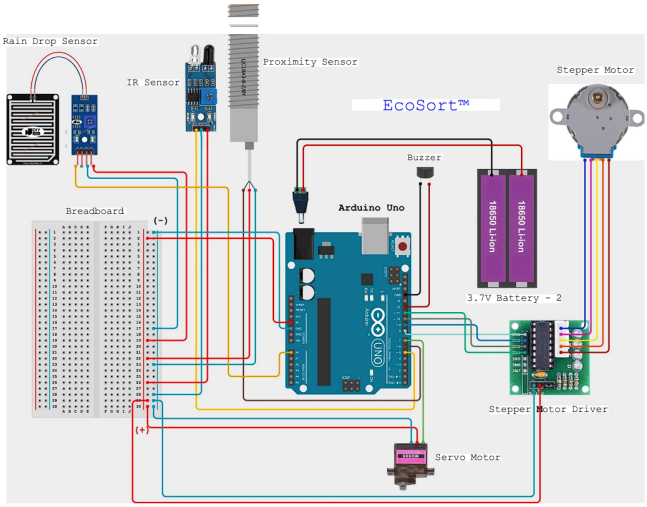


Fig. 1. Circuit Diagram

B. Waste Detection Mechanism

The EcoSort system uses a **hybrid waste detection mechanism** that combines **computer vision-based object detection with hardware sensor inputs** to improve classification accuracy. The system integrates a **YOLOv11 object detection model with multiple physical sensors**, allowing the system to perform reliable waste identification.

When a waste item is placed into the input slot, a **webcam captures an image of the object**. This image is processed by the **YOLOv11 model**, which analyzes visual features of the object and predicts its category. The model is trained to detect three waste categories:

- Metallic waste
- Wet waste
- Dry waste

The model outputs a **confidence score for each detected class**, indicating the probability that the object belongs to that category.

In addition to the vision-based detection, the system also incorporates several **hardware sensors** to capture physical characteristics of the waste. The hardware components used in the detection process include:

- **Arduino Uno microcontroller** – central control unit
- **Rain drop sensor** – detects moisture content to identify wet waste
- **Proximity sensor** – detects the presence of waste objects
- **IR sensor** – assists in object detection and positioning

The rain drop sensor helps determine whether the waste contains moisture, which is typically associated with **organic or wet waste**. The proximity and IR sensors confirm the presence and position of the object before detection begins.

To improve reliability, the EcoSort system applies a **weighted decision approach** that combines the results from both the YOLO model and the hardware sensors. Each detection method contributes a weighted score toward the final classification.

This **sensor fusion approach** improves the robustness of the EcoSort system by combining visual detection with physical sensing, reducing classification errors that may occur when using only a single detection method.



Fig. 2. Object Detection Results

C. Sorting Mechanism

Once the waste category is identified, the sorting mechanism directs the waste into the appropriate compartment.

The mechanical sorting system consists of a **rotating platform controlled by a stepper motor** and a **flap gate controlled by a servo motor**. The stepper motor rotates the internal platform to align the correct compartment with the waste entry point.

After alignment, the servo motor opens the flap gate, allowing the waste to fall into the designated compartment corresponding to the detected category.

This automated mechanism ensures that waste is properly segregated without requiring manual intervention.

IV. Analysis and Interpretation

The EcoSort prototype demonstrates the feasibility of using computer vision techniques for automated waste segregation. The YOLOv11 object detection model is capable of identifying different types of waste objects from images captured by the webcam.

The integration of machine learning with a motorized sorting mechanism enables the system to perform real-time waste

classification and segregation. The system successfully directs detected waste items into the appropriate compartments based on the model predictions.

Although the prototype shows promising results, the performance of the system depends on factors such as lighting conditions, camera placement, and the accuracy of the trained model. Mixed or partially visible waste objects may affect classification accuracy.

Despite these limitations, the EcoSort system highlights the potential of combining computer vision and embedded systems to create intelligent waste management solutions.

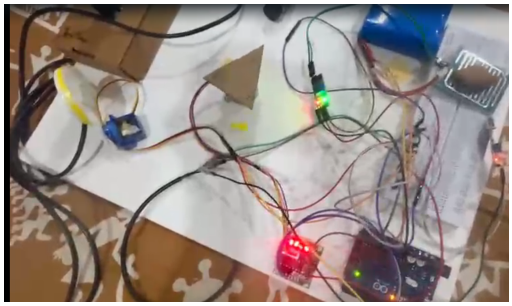


Fig. 3. Working Prototype

V. Conclusions and Recommendations

The EcoSort Smart Waste Management System presents an automated approach to waste segregation using **computer vision and machine learning technologies**. By integrating a webcam with the YOLOv11 object detection model, the system is capable of identifying different types of waste and directing them into appropriate compartments using a motorized sorting mechanism.

The proposed system reduces the need for manual waste segregation and improves the efficiency of recycling processes. The integration of machine learning techniques allows the system to identify waste based on visual features, providing greater flexibility compared to sensor-based methods.

Future improvements may include expanding the dataset to improve detection accuracy, integrating IoT connectivity for monitoring waste generation patterns, and optimizing the system for large-scale deployment in smart city environments.

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